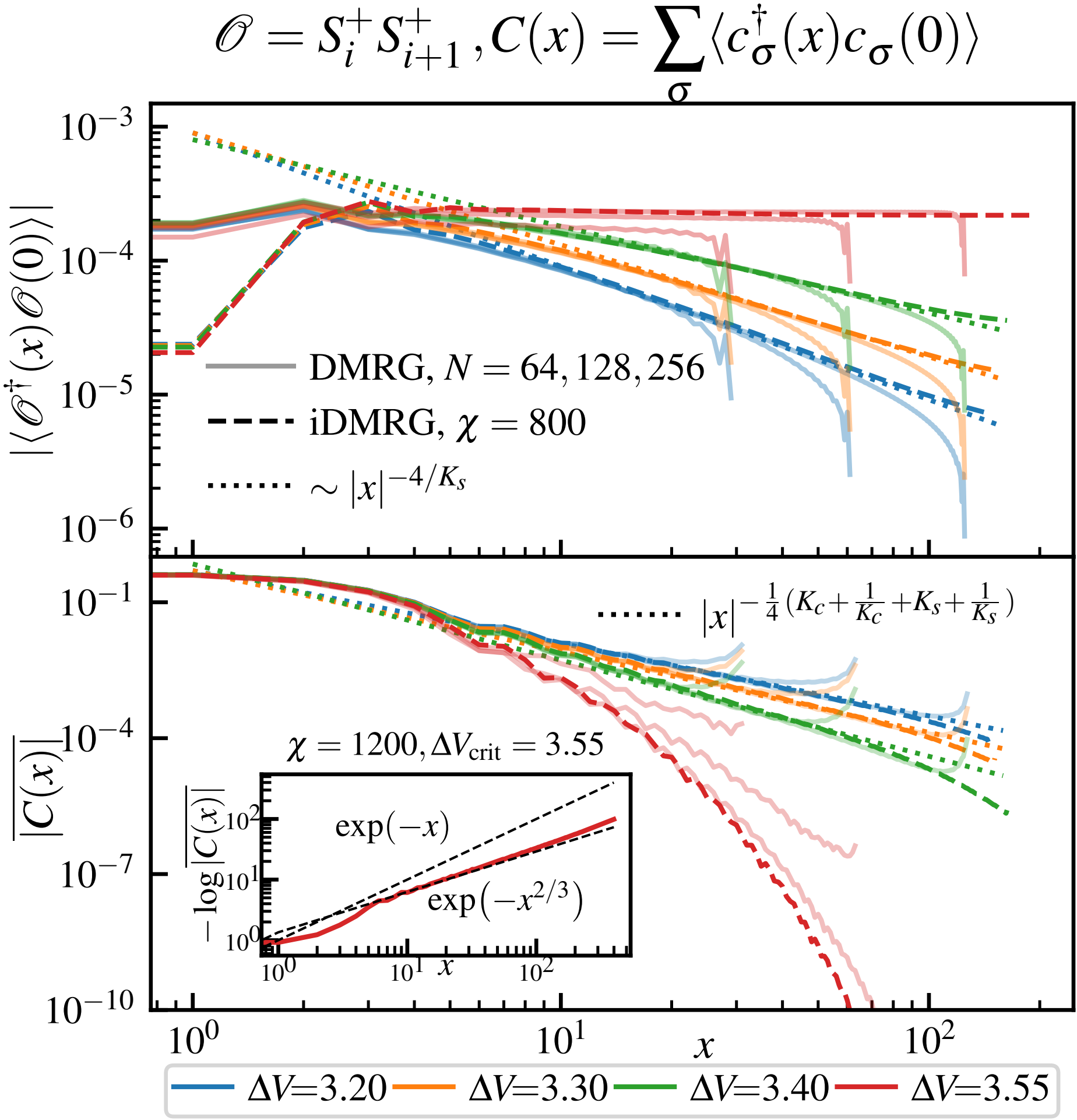

$\mathcal{O} = S_i^+ S_{i+1}^+, C(x) = \sum_\sigma \langle c_\sigma^\dagger(x) c_\sigma(0)\rangle$
$|\langle \mathcal{O}^\dagger(x)\mathcal{O}(0)\rangle|$
DMRG, $N = 64, 128, 256$
iDMRG, $\chi = 800$
$\sim |x|^{-4/K_s}$
$\overline{|C(x)|}$
$|x|^{-\frac{1}{4}(K_c+\frac{1}{K_c}+K_s+\frac{1}{K_s})}$
$\chi = 1200, \Delta V_{\text{crit}} = 3.55$
$-\log\overline{|C(x)|}$
$\exp(-x)$
$\exp(-x^{2/3})$
$x$
$\Delta V$=3.20
$\Delta V$=3.30
$\Delta V$=3.40
$\Delta V$=3.55